

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester III Mid-Term Examination, 2025-26 (NEP)

DEPARTMENT OF MATHEMATICS**Paper Code: MATMJ-31 : Linear Algebra (Mid-Term)**

Time: 1 Hour

Maximum Marks: 20

Instructions: Attempt any FIVE questions. All questions carry equal marks (4 marks each).

Question 1. Define a Vector Space over a field F . Prove that a non-empty subset $W \subseteq V$ is a subspace of V if and only if $\alpha u + \beta v \in W$ for all $\alpha, \beta \in F$ and $u, v \in W$. [4]

Question 2. Test the linear independence of vectors $v_1 = (1, 2, 1)$, $v_2 = (2, 1, 0)$, and $v_3 = (1, -1, 2)$ in $\mathbb{R}^3$. [4]

Question 3. State and prove the Rank-Nullity Theorem for a linear transformation $T: V \rightarrow W$. [4]

Question 4. Find the basis and dimension of the subspace $W = \{(x, y, z) \in \mathbb{R}^3 \mid x + 2y - z = 0\}$. [4]

Question 5. Let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be defined by $T(x, y) = (2x + y, x - 3y)$. Find the matrix of T relative to the standard basis of $\mathbb{R}^2$. [4]

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